

altairsim — Altair 4K BASIC off a cassette

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```
altairsim basic4k.toml

MEMORY SIZE? <return>
TERMINAL WIDTH? <return>
WANT SIN? Y

ALTAIR BASIC VERSION 3.1
[FOUR-K VERSION]
742 BYTES FREE
OK
```

Altair 4K BASIC 3.1, read off a period cassette image by the bootstrap MITS shipped, both unmodified. The machine is the bare 1975 Altair: 4K of RAM, an 88-2SIO for the Teletype, an 88-ACR for the recorder, sense switches at `0x80`.

The three lines in `basic4k.toml`'s `startup` are the three things a human did in 1975 — put the tape in and press PLAY (`MOUNT`), toggle in the bootstrap (`LOAD`), and run it from zero.

The files

File	What it is
<code>basic4k.toml</code>	The operator. <code>base = "basic4k"</code> is the hardware (see <code>machines/basic4k.toml</code>); this file puts the tape in the recorder.
<code>4K BASIC Ver 3-1.tap</code>	The cassette.
<code>LDR4K31.HEX</code>	The period bootstrap, assembled.

The tape comes off in about a second rather than the 110 the real 300-baud cassette took, because the default clock is flat out. `SET cpu0 clock_hz=2000000` buys back the 2 MHz machine **and** its 110 seconds — BASIC cannot tell the difference either way.

`LDR4K31.ASM` and its listing are in `tapes/4KBasic31/`: source rather than product, so they stay in the repository and are not in the package.